Software Requirements Specification

for Online Exam Portal

Version 1.1 Draft

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Revision History

Name	Date	Reason For Changes	Version
Darshana S	7-09-2026	Initial draft (11 requirements: 7 functional, 4 non-functional)	1.0 Draft
Darshana S	23-10-2026	Updated Appendix C (Requirement Traceability Matrix) with Test Case IDs mapped to FR-1–FR-7 and NFR-1–NFR-4, as documented in the Test Plan.	1.1 Draft

1. Introduction

1.1 Purpose

This SRS specifies the requirements for the Online Exam Portal, version 1.0. It covers the complete web application: user accounts, MCQ and exam creation, timed exam delivery, automatic evaluation and result viewing.

1.2 Intended Audience and Reading Suggestions

- Developers (Arati, Prakash): Sections 3, 5 and 6.
- Project manager (Chathurth): Sections 1, 2 and 5.
- Testers and documentation (Darshana): Sections 5 and 6 and Appendix C. Test cases are derived from the requirement IDs (FR-1 to FR-7, NFR-1 to NFR-4).
- Instructors and faculty reviewers: Sections 1 and 2.

Section 2 gives the overall description, Section 3 the interfaces, Section 4 the analysis models, Section 5 the functional requirements, and Section 6 the non-functional requirements.

1.3 Product Scope

The Online Exam Portal is a web-based system where instructors create multiple-choice exams and students take them within a timed window. The system grades responses automatically and shows results. Objectives:

- Replace paper-based MCQ tests with an online process.
- Give instant, accurate scoring.
- Give instructors a simple view of student performance.

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
- Course-provided SRS and Test Plan templates, PES University
- Django documentation, https://docs.djangoproject.com

2. Overall Description

2.1 Product Perspective

The Online Exam Portal is a new, self-contained web application. A Django application serves the pages and stores data in a SQLite database. Users access it through a web browser.

2.2 Product Functions

- Registration, login and role-based access (Student, Instructor)
- MCQ question management
- Exam creation and scheduling
- Timed exam attempt and submission
- Automatic evaluation and scoring
- Result viewing for students and class reports for instructors

- User management through the built-in Django admin site (not a separate requirement)

2.3 User Classes and Characteristics

User Class	Description	Importance
Student	Takes exams and views own results; limited technical expertise	High
Instructor	Creates questions and exams, views class results; moderate technical expertise	High
Administrator	Manages accounts through the Django admin site; created by the developers, not through registration	Low

2.4 Operating Environment

- Client: current versions of Chrome and Firefox on Windows, Linux or macOS.
- Server: Python 3.11+ and Django 5.x on Windows or Linux.
- Database: SQLite.

2.5 Design and Implementation Constraints

- Backend in Python (Django); pages served by the Django application using HTML, CSS and JavaScript. JavaScript is used for the countdown timer and question navigation.
- Authentication uses Django sessions.
- Only multiple-choice questions with one correct answer are supported in v1.0.
- Development follows the semester timeline and uses the team's public GitHub repository.

2.6 Assumptions and Dependencies

- Students have a stable internet connection during exams.
- The server clock is the authoritative time source for all timers.
- All students may see every published exam; per-student enrollment is out of scope for v1.0.

3. External Interface Requirements

3.1 User Interfaces

- Screens: Register, Login, Student Dashboard, Exam Instructions, Exam Attempt (question, options, navigator, countdown timer, Previous/Next/Submit), Result, Instructor Dashboard, Question Form, Exam Form, Class Report.
- Every screen has a header with the user name and a Logout button.
- Validation errors appear next to the field they belong to.

3.2 Software Interfaces

- Database: SQLite accessed through the Django ORM.
- Browser: standard HTML forms and JavaScript.

3.3 Communications Interfaces

- HTTP between browser and server (localhost during development; HTTPS if deployed).

4. Analysis Models

4.1 Use Case Diagram

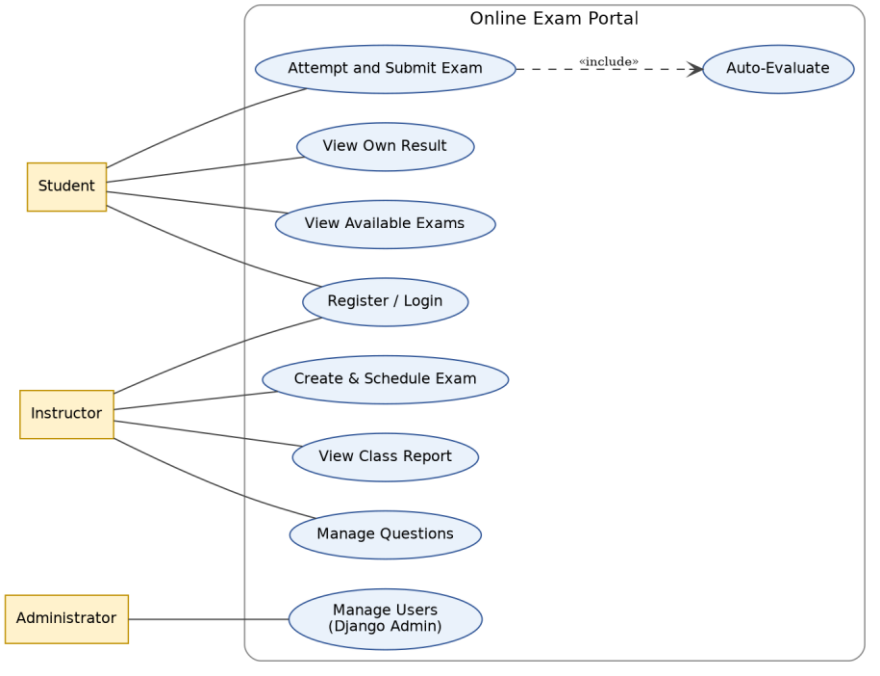


Figure 1: Use case diagram

4.2 Entity-Relationship Diagram

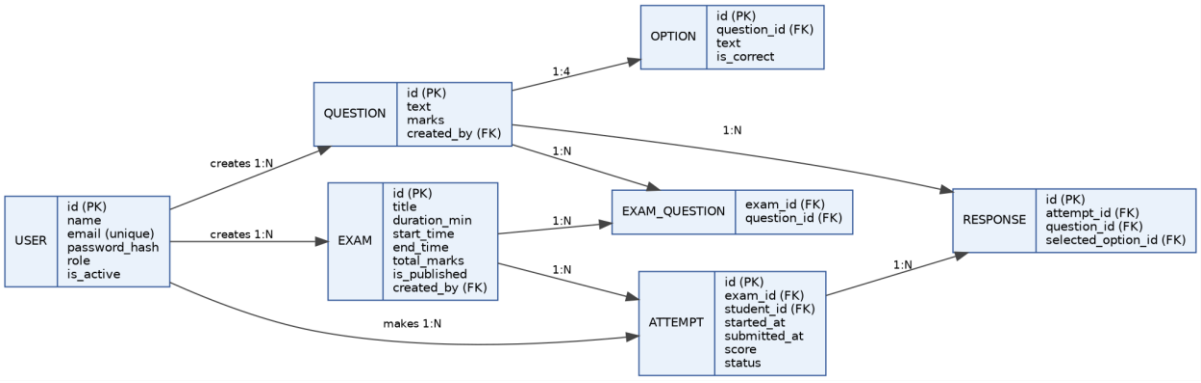


Figure 2: Entity-relationship diagram

5. System Features

5.1 User Authentication

5.1.1 Description and Priority

High priority. Users create accounts and log in; the system restricts pages by role.

5.1.2 Stimulus/Response Sequences

- User submits the registration form: system validates, creates the account and redirects to Login with a confirmation.
- User submits credentials: system authenticates and opens the dashboard for the role.
- Invalid credentials: system shows "Invalid email or password".

5.1.3 Functional Requirements

FR-1: User Registration

The system shall allow a new user to register as a Student or an Instructor.

- The form contains full name, email, password, confirm password and role; all fields are mandatory.
- The email must be in a valid format and not already registered.
- The password must have at least 8 characters with at least one letter and one digit.
- Password and confirm password must match.
- Each invalid field shows a message next to it, and no account is created.
- On success the account is created, the password is stored hashed, and the user is redirected to Login with a confirmation message.
- Administrator accounts cannot be created through registration.

FR-2: Login, Logout and Role-Based Access

The system shall authenticate users and restrict pages according to their role.

- Users log in with email and password and are redirected to the Student or Instructor dashboard.
- Wrong credentials show "Invalid email or password" without saying which one was wrong.
- Logout ends the session; the browser Back button then cannot open protected pages.
- Unauthenticated users opening a protected page are redirected to Login.
- A Student opening an instructor page, or an Instructor opening a student page, sees an access-denied (403) message.
- Deactivated accounts cannot log in.
- A session expires after 30 minutes of inactivity.

5.2 Exam Management

5.2.1 Description and Priority

High priority. Instructors build a bank of MCQs and use it to create scheduled exams.

5.2.2 Stimulus/Response Sequences

- Instructor submits a question form: system validates and saves it.
- Instructor submits an exam form with selected questions: system saves it as a draft.
- Instructor publishes the exam: it becomes visible to students.

5.2.3 Functional Requirements

FR-3: Create and Manage MCQ Questions

The system shall allow an Instructor to create, edit, delete and list their own multiple-choice questions.

- A question has text, exactly 4 options, exactly 1 correct option, and marks (a positive integer, default 1).
- The system rejects empty question text, any blank option, duplicate options, and a question with no correct option.
- An Instructor sees, edits and deletes only their own questions.
- A question used in an exam that already has attempts cannot be edited or deleted.
- Deleting a question shows a confirmation and removes it from draft exams.

FR-4: Create and Schedule an Exam

The system shall allow an Instructor to create an exam from their questions, set its schedule and publish it.

- An exam has a title, duration in minutes (1 to 180), start date-time, end date-time, and at least one selected question.
- The end time must be after the start time, and the start time cannot be in the past when creating.
- The duration cannot exceed the length of the window between start and end time.
- Total marks are calculated automatically as the sum of the selected questions' marks.
- An exam stays a draft until published; only published exams are visible to students.
- An exam with attempts cannot be edited or deleted.
- An Instructor sees, edits and deletes only their own exams.

5.3 Exam Taking and Evaluation

5.3.1 Description and Priority

High priority. Students take a published exam inside its window; the system grades the attempt automatically.

5.3.2 Stimulus/Response Sequences

- Student opens the dashboard: system lists published exams with their status.
- Student starts an open exam: system creates the attempt and starts the timer.
- Student submits, or the timer reaches zero: system locks the attempt, scores it and shows the result.

5.3.3 Functional Requirements

FR-5: Attempt an Exam

The system shall let a Student view available exams and attempt an open exam under a timer.

- The Student dashboard lists published exams with title, start, end, duration and status (Upcoming, Open, Closed, Completed).
- A Student can start an exam only while it is Open, that is between its start and end time.
- Each Student can attempt an exam only once; a second start attempt is refused with a message.
- An instructions page shows duration, number of questions and total marks before the exam starts.
- The attempt page shows one question at a time with 4 options, Previous/Next buttons and a question navigator marking answered questions.

- A countdown timer runs from the server-side start time; the attempt ends at the duration or at the end of the exam window, whichever is earlier.
- A Student can change answers before submitting and can leave questions unanswered.

FR-6: Submit an Exam and Automatic Evaluation

The system shall accept the submission of an attempt and evaluate it automatically.

- Manual submit asks for confirmation and shows the number of unanswered questions.
- When the timer reaches zero the attempt is submitted automatically with the answers selected so far.
- After submission the attempt is locked; Back or reload cannot change answers.
- The server rejects any answer received after the deadline.
- Each correct answer earns the question's full marks; wrong and unanswered questions earn zero (no negative marking).
- The total score and percentage are stored and calculated exactly once, at submission.
- The result page opens immediately after submission.

5.4 Results

5.4.1 Description and Priority

Medium priority. Students see their own results; Instructors see how a class performed.

5.4.2 Stimulus/Response Sequences

- Student opens My Results: system lists their attempts with scores.
- Instructor opens an exam report: system lists all attempts with summary figures.

5.4.3 Functional Requirements

FR-7: View Results

The system shall show a Student their own results and an Instructor the results of their own exams.

- A Student sees score, total marks, percentage and a Pass/Fail status (pass mark 40%) for each submitted attempt.
- A Student can list all their past attempts with dates and scores.
- A per-question review (own answer against correct answer) becomes available only after the exam window closes.
- No result is shown for an exam that has not been submitted.
- An Instructor sees a report per exam listing each student's name, email, score, percentage and submission time.
- The report shows the number of attempts and the average, highest and lowest score.
- A Student cannot see other students' results, and an Instructor cannot see reports of other instructors' exams.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

NFR-1: Performance

The system shall respond quickly on the demonstration setup.

- Normal pages load within 3 seconds on the demonstration machine.
- Evaluation finishes and the result page opens within 2 seconds of submission for exams of up to 50 questions.
- The system supports at least 10 students taking an exam at the same time.

6.2 Safety Requirements

NFR-2: Safety (Data Protection)

The system shall protect exam data from loss or unintended change.

- A submitted attempt is saved to the database before its result is shown, so a later server restart does not lose it.
- Submitted attempts and scores cannot be edited or deleted by Students or Instructors through the application.
- A question or exam that has attempts cannot be deleted, so past results are never orphaned.
- If saving a submission fails, the system shows an error and keeps the attempt open so the Student can retry before the deadline.

6.3 Security Requirements

NFR-3: Security

The system shall protect accounts and exam data.

- Passwords are stored only as salted hashes (Django default hashing).
- Role checks are enforced on the server for every request, not only by hiding menu items.
- All forms are protected against CSRF, and database access uses the Django ORM (no raw SQL with user input).
- Correct answers are never sent to the browser while an exam is in progress.
- Protected pages require a valid session.

6.4 Software Quality Attributes

NFR-4: Usability and Compatibility

The system shall be easy to use and work in common browsers.

- Works in current versions of Chrome and Firefox with no plugins.
- Layouts remain usable from 360 px screen width upward.
- A Student can open and start an available exam within 3 clicks of logging in.
- Error and confirmation messages use a consistent style and wording on every screen.

6.5 Business Rules

- Only Instructors create questions and exams; only Students attempt exams.
- Each Student has exactly one attempt per exam.

- The pass mark is 40% of the total marks.
- Instructors access only their own questions, exams and reports.

Domain requirements: Exam timing always uses the server clock. Question marks are positive integers. An attempt is either In Progress or Submitted.

7. Other Requirements

- Database migrations are kept under version control in the repository.
- The SQLite database file is backed up before each demonstration (a team practice, not a tested software requirement).
- The code and all project documents live in the team's public GitHub repository.

Appendix A: Glossary

Term	Meaning
MCQ	Multiple-Choice Question
Attempt	One student's single sitting of an exam
Exam window	Time between the exam start and end date-times in which it can be started
FR / NFR	Functional / Non-functional Requirement
CSRF	Cross-Site Request Forgery

Appendix B: Field Layouts

Entity	Fields
User	id, name, email (unique), password_hash, role, is_active
Question	id, text, marks, created_by
Option	id, question_id, text, is_correct
Exam	id, title, duration_min, start_time, end_time, total_marks, is_published, created_by
ExamQuestion	exam_id, question_id
Attempt	id, exam_id, student_id, started_at, submitted_at, score, status
Response	id, attempt_id, question_id, selected_option_id

Appendix C: Requirement Traceability Matrix

Req ID	Requirement	Section	Priority	Test Case IDs
FR-1	User Registration	5.1	High	UT_01, UT_02, UT_03, UT_04, UT_05, UT_06, UT_07, UT_08, IT_01, ST_01, UT_44
FR-2	Login, Logout and Role-Based Access	5.1	High	UT_09, UT_10, UT_11, UT_12, UT_13, IT_02, IT_03, ST_02, ST_03, ST_04

FR-3	Create and Manage MCQ Questions	5.2	High	UT_14, UT_15, UT_16, UT_17, UT_18, UT_19, UT_20, IT_04, IT_05, ST_05, UT_45
FR-4	Create and Schedule an Exam	5.2	High	UT_21, UT_22, UT_23, UT_24, UT_25, UT_26, IT_06, IT_07, ST_06, UT_46
FR-5	Attempt an Exam	5.3	High	UT_27, UT_28, UT_29, UT_30, ST_07, IT_08, UT_31, UT_32, ST_08, IT_09
FR-6	Submit an Exam and Automatic Evaluation	5.3	High	UT_33, IT_10, ST_09, ST_10, UT_34, UT_35, IT_11, ST_11
FR-7	View Results	5.4	Medium	UT_36, UT_37, ST_12, UT_38, IT_12, UT_39, ST_13, ST_14
NFR-1	Performance	6.1	Medium	PT_01, PT_02, PT_03
NFR-2	Safety (Data Protection)	6.2	Medium	ST_15, ST_16, ST_17
NFR-3	Security	6.3	High	ST_18, ST_19, ST_20, ST_21, UT_47
NFR-4	Usability and Compatibility	6.4	Medium	UT_40, UT_41, UT_42, UT_43